Isoperimetric Inequalities on Matroid

Zejia Chen

September 30, 2026

Introduction. Isoperimetric inequalities, which relate the area to the perimeter of its boundary, have been widly studied in many areas of mathematics. For example, in the two dimensional Euclidean space, the classical isoperimetric inequality states that for closed curve with length L, the area A of the region that it encloses satisifes: $4\pi A \leq L^2$.

Instead of considering the continuous space, we foucs on the isoperimetric inequalities on Boolean hypercube $\Omega = \{\pm 1\}^n$ with the distribution μ over Ω. For a subset $S \subseteq \Omega$, the analogue of its volume is measured by $\mu(S)$. In the following, we represent a subset $S \subseteq \Omega$ by a Boolean function $f : \Omega \mapsto \{\pm 1\}$. In this notation, $\mathrm{Var}_\mu[f] = 4\mu(S)(1-\mu(S)) \geq 2\min\{\mu(S), 1-\mu(S)\}$ serves as the volume term, measuring the smaller side of the partition in terms of the fixed boundary. There are several natural notions of boundary on the Boolean hypercube. Two classical ones are based on the *influence* and the *sensitivity* of coordinates. We introduce the influence of i-th coordinate and the sensitivity with respect to the boolean function f as follows:

$$\mathsf{Inf}_{\mu,i}(f) = \Pr_\mu[f(x) \neq f(x_i)] \quad \text{and} \quad s_\mu(S)(x) = \sum_{i=1}^n \mathbb{1}[f(x) \neq f(x^i)],$$

where x^i is obtained by filpping the sign of i-th coordinate of x. These quantities have a natural graph correspondance with respect to the n-dimensional hypercube graph. For subset of vertices $S = \{f(x) = 1 \mid x \in \Omega\}$, both influence and sensitivity quantify the size of the edge boundary $\partial S = \{(i, x) : f(x) \neq f(x^i)\}$ from different perspectives.

One fundamental isoperimetric inequality is the Poincaré inequality: for boolean function f, $2\min\{\mu(S), 1-\mu(S)\} \leq \mathrm{Var}_\mu[f] \leq \sum_{i=1}^n \mathsf{Inf}_{\mu,i}(f)$. In this work, we focus primally the following isoperimetric inequalities, for any Boolean function $f : \Omega \mapsto \{\pm 1\}$:

1. **Talagrand's L^1–L^2 inequality** [Tal94]: $\mathrm{Var}_\mu[f] \lesssim \sum_{i=1}^n \dfrac{\mathsf{Inf}_{\mu,i}(f)}{1 + \log(1/\mathsf{Inf}_{\mu,i}(f))}$.

2. **Eldan–Gross inequality** [EG22]: $\mathrm{Var}_\mu[f] \cdot \sqrt{\log\left(2 + \dfrac{e}{\sum_{i=1}^n \mathsf{Inf}_{\mu,i}(f)^2}\right)} \lesssim \mathbb{E}\left[\sqrt{s(f)}\right]$.

Talagrand's inequality provides a logarithmic improvement over the Poincaré inequality when influences are small, while the Eldan–Gross inequality relates the square-root sensitivity boundary to influences across coordinates. These inequalities have a wide range of appliations, such as voting models with correlated preferences [KLMM24].

Most previous work presents isoperimetric inequalities equipped with the fully-support measure on the Boolean hypercube under suitable weak-dependence assumptions. By contrast, analogous inequalities on classical combinatorial structures, such as matroids and high-dimensional expanders, still remain unexplored, even for element-weighted distributions. This motivates the following conjecture for this proposal:

Conjecture 0.1. *The Talagrand's L^1–L^2 inequality and Eldan-Gross inequality hold for element-weighted distribtion over matroid independent sets and bases.*

Approach and Previous Work. Talagrand's L^1–L^2 inequality was first proved for biased product distribution [Tal94]. It was later extended to Markov random fields on bounded-degree graphs under suitable hypercontractivity assumptions [KLMM24]. Eldan and Gross proved their inequality for the uniform measure using pathwise stochastic analysis [EG22]. Ivanisvili and Zhang subsequently gave a simpler semigroup proof that also applies to biased product measures [IZ26]. In my recent work with Zongchen Chen and Xinyuan Zhang [CCZ26], we established the isoperimetric inequalities in terms of the weak correlation property of the underlying distribution through the Markov semigroup and gradient estimates.

A *Markov semigroup* $(P_t)_{t\geq 0}$ describes the time evolution of a Markov process and is generated by an operator L ($P_t = e^{tL}$): for any $f : \Omega \mapsto \mathbb{R}$ and $x \in \Omega$, $Lf(x) = \sum_{i=1}^{n} q_i(x)(f(x^i) - f(x))$. Here $\{q_i\}_{i\in[n]}$ is called the *transition rates* of the Markov semigroup, which is always assumed to be *reversible* with respect to μ: $\mu(x)q_i(x) = \mu(x^i)q_i(x^i)$. Given the Markov semigroup $(P_t)_{t\geq 0}$, the corresponding strong gradient estimate holds for constant ρ if for any function $f : \Omega \mapsto \mathbb{R}$ and $t \geq 0$,

$$\|\nabla P_t f\|_2 \leq e^{-\rho t} P_t \left(\|\nabla f\|_2\right),$$

where the discrete gradient of f is defined as $\nabla f(x) = [\sqrt{q_i(x)}(f(x^i) - f(x))]_{i\in[n]}$. Through the potential function and derivative techniques, the gradient estimate can be equivalently written as a functional inequality, which is known as the Bakry–Émery-type condition [BÉ85]. This can be formulated as a local matrix eigenvalue problem [CKLP22].

Previous work established the implicition between the gradient estimates and Talagrand's L^1–L^2 inequality under the assumption of hypercontractivity [CL12]. In [CCZ26], this implicition was strengthened to Eland–Gross's inequality using a stronger, coordinate-wise gradient estimate and the semigroup framework of [IZ26]. By analyzing commutators between the generator and discrete gradient ($(L\nabla - \nabla L)f = H \cdot \nabla f$), we also derived sufficient local matrix criteria for these estimates and verified them under suitable weak-dependence assumptions on μ.

Future Work.

References

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